SHRIKANT SHARMA

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Data Scientist | Applied AI, LLM Evaluation & Machine Learning | Python, SQL, LangGraph, AWS

Data Scientist with ~8 years across regulated pharma and financial services, combining production machine learning with recent depth in agentic AI and LLM evaluation. Ships models end to end across 3 live systems: benchmarked model selection, calibration, deployment as monitored services, and drift detection. Builds the evaluation frameworks that prove a system improved (LLM-as-judge, faithfulness, regression suites), and applies causal inference to separate real effects from confounding.

SKILLS

Machine Learning & AI: LLM Evaluation (LLM-as-a-Judge, Faithfulness, precision@k, MRR), Agentic AI, RAG & Vector Search, Causal Inference (PSM, G-Computation, IPW), Experiment Design & A/B Testing, MLOps, Drift Monitoring & Observability, Feature Engineering, Guardrails, Prompt Engineering, Model Governance, NLP, Gradient Boosting, Deep Learning, Anomaly Detection, Segmentation, Time-to-Event Forecasting, Calibration, Explainability (SHAP)

Languages & Libraries: Python (Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, Hugging Face Sentence Transformers, FAISS, SHAP, RAGAS, LangGraph, DoWhy, lifelines, PySpark, Matplotlib), SQL, R

Cloud & Data Platforms: AWS (S3, EC2, Lambda, ECR), Databricks, Snowflake

Tools & Platforms: MLflow, Docker, Git, FastAPI, Apache Spark, Claude Code, Model Context Protocol (MCP), Streamlit, Tableau, Power BI

EXPERIENCE

Amgen | Data Scientist **Sept 2022 – May 2026**

- Led an initiative to apply NLP to unstructured compliance documents, developing and deploying a text classification model (TF-IDF and Random Forest in Python, Scikit-learn), engineering the feature pipeline and evaluating classifier performance on held-out data before rollout; cut manual document categorization by 25%.
- Prototyped LLM applications across 2 company-wide GenAI hackathons covering audit finding summarization and question-answering over regulatory documents, evaluating output faithfulness against source material; findings informed the enterprise GenAI roadmap.
- Designed a multivariate statistical risk model in Python (Pandas, NumPy, Scikit-learn) scoring 65+ ex-US countries and 40+ US products across 20+ indicators. Rebalanced weights from 50/50 to 75/25 toward quantitative signals and expanded output from 3 tiers to 5, reordering the risk ranking and surfacing 3 high-risk markets that had gone undetected. Detected model drift across annual cycles by comparing score distributions, then rebuilt and revalidated. Presented methodology to Directors and Executive Directors, informing resource allocation across 3 regions.
- Implemented statistical anomaly detection for transaction monitoring across 5 payment processes on 50K+ records (US, ELMAC, JAPAC), using peer-group z-scores with tuned thresholds and flag-rate stability tracking to catch drift; reduced manual review volume by 35%.
- Architected production data pipelines on AWS and Databricks behind the Amgen CMS Open Payments federal submission (~509K records, ~$630M), with automated SQL validation gates reconciling record counts from ingestion through submission so every rejected transaction was accounted for, holding the failure rate under 0.5% across 4 annual cycles. Created year-over-year Tableau dashboards and automation that cut cycle time from 2 weeks to 4 days.
- Mentored 3 junior data scientists on Python automation, modeling practice, and reproducibility, and authored analytics workflow documentation adopted as team reference; delivered 3 concurrent projects on schedule.

Nugget.ai | Data Scientist Intern **Jan 2022 – May 2022**

- Trained models in Python (Scikit-learn) to predict candidate-role fit from behavioral assessment data, engineering 20+ features spanning structural, linguistic, behavioral, and content-based signals. Compared Logistic Regression, Random Forest, and Gradient Boosting with cross-validation, improving matching accuracy 15% over the platform baseline.

Atos Syntel | Data Scientist | *Client: American Express* **Feb 2018 – Dec 2019**

- Engineered production risk scoring pipelines in Python and SQL over 10M+ transactions, deriving behavioral features around spending velocity, geographic patterns, and customer segmentation that fed fraud and credit risk decisioning and weekly review.
- Designed Tableau dashboards consolidating 5+ source systems into a single executive view used in the weekly credit risk review, supporting mitigation decisions totaling over $2M and cutting reporting time by 30%.
- Owned the A/B testing framework in Python, SQL, and Git for credit risk strategy experiments, handling experiment design, power-based sample sizing, and significance testing with multiple-testing correction across 12+ cycles per year.

Atos Syntel | Data Analyst | *Client: American Express* **Mar 2016 – Jan 2018**

- Developed the SQL data pipelines processing 5M+ financial transaction records monthly, with automated validation logic and data quality checks that held accuracy at 99.5% for regulatory reporting.
- Automated 15+ recurring reports using SQL and Python, freeing 30+ hours of manual work each month for analysis instead of data prep.

APPLIED AI & MACHINE LEARNING PROJECTS

Clinical Trial Intelligence System | *Python*, *LangGraph*, *Sentence Transformers*, *PubMedBERT*, *Cross-Encoder*, *Groq Llama 3.3 70B*, *FAISS*, *RAGAS*, *Streamlit*, *Git* **May 2026**

github.com/Shrikant-Sharma/clinical-trial-rag | shrikant-clinical-rag.streamlit.app

- Designed and deployed an agentic Corrective RAG system on LangGraph over 484 ClinicalTrials.gov oncology protocols (3,264 chunks): multi-step workflow with LLM-as-judge grading, query rewriting, conditional routing, and bounded retries. 3 independent guardrails reject adversarial queries scoring 0.91 cosine similarity that a single confidence threshold missed.
- Created the evaluation harness that measures it: a curated 25-query golden set spanning in-domain and adversarial cases, RAGAS faithfulness scoring, retrieval metrics (precision@k, MRR), and an LLM-judged regression suite rerun after every change; the harness quantified a 34-point refusal-rate reduction from a single architectural change.
- Engineered the retrieval stack from measured failure: compared 3 chunking strategies, switched to domain-tuned PubMedBERT embeddings after a general-purpose model scored relevant oncology documents at 0.24 similarity, and added a cross-encoder reranker for 2-stage retrieval, halving in-corpus refusal rate (67% to 33%). Deployed live on Streamlit with graceful rate-limit degradation.

Patient Risk Stratification Pipeline | *Python (Scikit-learn*, *XGBoost*, *PyTorch*, *SHAP*, *DoWhy*, *lifelines*, *scikit-survival)*, *MLflow*, *FastAPI*, *Docker*, *AWS (Lambda*, *ECR*, *Mangum)*, *Git* **May 2026**

github.com/Shrikant-Sharma/patient-risk-stratification | Live API on AWS Lambda

- Delivered an end-to-end ML pipeline for clinical heart disease risk stratification on UCI Heart Disease (303 patients, 13 clinical features); selected XGBoost from 5 model families via MLflow experiment tracking and 5-fold cross-validation, reaching 0.95 AUC-ROC and 0.93 recall (precision 0.81) at the clinical operating threshold.
- Added SHAP TreeExplainer for global and per-patient feature attributions critical for clinical decision support, validated probability calibration at Brier 0.092 with reliability curves, and tuned the decision threshold to prioritize recall while maintaining precision >= 0.80.
- Extended the pipeline to causal inference and survival analysis on the NHEFS cohort (1,629 subjects); estimated treatment effects via propensity score matching and G-computation with DoWhy (3-way convergence within 0.2 kg of the IPW reference, 3 sensitivity tests passing) and modeled 10-year mortality with Cox PH and Random Survival Forest at 0.80 concordance, revealing that unadjusted log-rank significance (p=0.005) was pure age confounding.
- Deployed live as a public REST API on AWS Lambda via Amazon ECR and Mangum; multi-stage Docker build, EventBridge cold-start warmer holding warm latency at ~275ms, FastAPI Swagger UI, and monitoring of input and prediction distributions for drift.

Payment Transaction Anomaly Detection | *Python (Pandas*, *Scikit-learn*, *PySpark)*, *Snowflake*, *Tableau*, *Git* **May 2026**

github.com/Shrikant-Sharma/pharma-compliance-spend-analytics | public.tableau.com/app/profile/shrikant.sharma

- Developed a payment-transaction anomaly detection system on 16M+ real payment records ($13B) for large-scale transaction-transparency review, performing entity resolution across 989K sampled transactions to 289K counterparties and engineering 5 entity-level features.
- Applied within-segment z-scores plus global IQR with a monetary floor and concentration logic, flagging the top 1.67% of entities as a HIGH-risk triage queue.
- Layered Isolation Forest anomaly detection on top; reconciliation against the rule-based flags surfaced 3 distinct anomaly archetypes (caught by BOTH methods, rules-only, and model-only), revealing suspicious patterns no single method catches alone.
- Engineered the same feature pipeline across 3 execution backends (Pandas, Snowflake, PySpark) with row-level equivalence verified; Snowflake SQL surfaced that the top 4 counterparties concentrate 59.7% of segment payments, visualized in 2 interactive Tableau dashboards.

EDUCATION

Clark University, Worcester, MA | Master of Science in Data Analytics **May 2022**

Sant Gadge Baba Amravati University, India | B.E. in Electronics & Telecommunication **Feb 2016**

ACHIEVEMENTS

Acclaim Award (3-Star) | Amgen | For data science delivery and cross-functional collaboration **Feb 2025**